GL-ProQ-en-016

DFMEA Procedure

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Purpose	The purpose of this procedure is to provide guidelines that will enable a design team to conduct and complete a successful and effective Design Failure Mode and Effects Analysis (DFMEA). The DFMEA is a systematic design risk management methodology intended to identify the offer's / solution's *design weaknesses, and mitigate those potential design risks* through a *proactive*, *qualitative*, *structured*, *inductive* & *documented* reasoning.
Scope	This procedure applies to all Schneider Electric **Offers** at any level of hierarchy (System, Sub-system, or Component) covering various design aspects, including Hardware, Software and Firmware, projects and services.

Target Audience

- Offer Technical Leaders
- Design Engineers
- Offer Safety Officers
- Offer Quality Leaders
- DfSR Leaders and Practitioners
- Marketing Leaders, League Leaders, Product Owners
- Scrum Masters

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General

References

- **GL-DirQ-en-044 Design for Safety & Reliability (DfSR) Directive**
- **GL-ProQ-en-019 DfSR Plan Procedure**
- **GL-ProQ-en-024 DfSR Concept Selection Procedure**
- **GL-ProQ-en-025 P-Diagram Procedure**
- **GL-ProQ-en-018 Fault Tree Analysis (FTA) Procedure**
- **GL-ProQ-en-029 DfSR Mission Profile Procedure**
- **GL-ProQ-en-030 DfSR Requirements & Allocation Procedure**
- **GL-ProQ-en-092 Offer Safety & Regulations Review Procedure**
- **GL-DirQ-en-025 Offer Safety Process**
- **GL-DirQ-en-013 Managing Offer Safety Risks**

All listed references are part of the global DfSR framework and need to be considered as a background while proceeding to DFMEA.

Process Inputs

To conduct a given DFMEA, the team lead by a DFMEA Facilitator, should prepare/collect the following:

1. Marketing Requirements (MKT-005)
2. DfSR plan
3. Historical Quality Performance (Lessons Learned)
4. Functional Analysis & Functional Requirements
5. System Architecture
6. Available P-Diagram(s) and DFMEA(s) of similar product(s)
7. Developed P-diagrams on priority S/S/C's with CTQ inputs from QFD matrices (see **GL-ProQ-en-025 P-Diagram Procedure**)
8. Field Reliability and Safety History of similar product(s) (if available)
9. Design knowledge (e.g., VOC/CTQs, Failure Analysis / PRB Reports, Schematics, Drawings, … etc.) of the offer under analysis and similar ones that are in the field.

Important: Reliasoft XFMEA module is the required software for DFMEA development

Process Overview

DFMEA teams should follow the 3 steps to ensure successful execution of a DFMEA:

A. **Conducting the DFMEA in XFMEA module of Reliasoft**
B. **Utilizing the Results of the DFMEA in driving other DfSR Tasks**
C. **Sustaining the DFMEA**

A. Conducting the DFMEA

The DFMEA for the new offers / solutions MUST be developed using XFMEA module of the Reliasoft Software by using already developed p-diagram. Below is the step-by-step explanation of the DFMEA fields on the electronic form in the order of execution:

1. FUNCTION:

Begin the DFMEA, using p-diagram, enter the functions from p-diagram into the Functions area. Please refer to **GL-ProQ-en-025 P-Diagram Procedure** for guidance of building the P-Diagram.

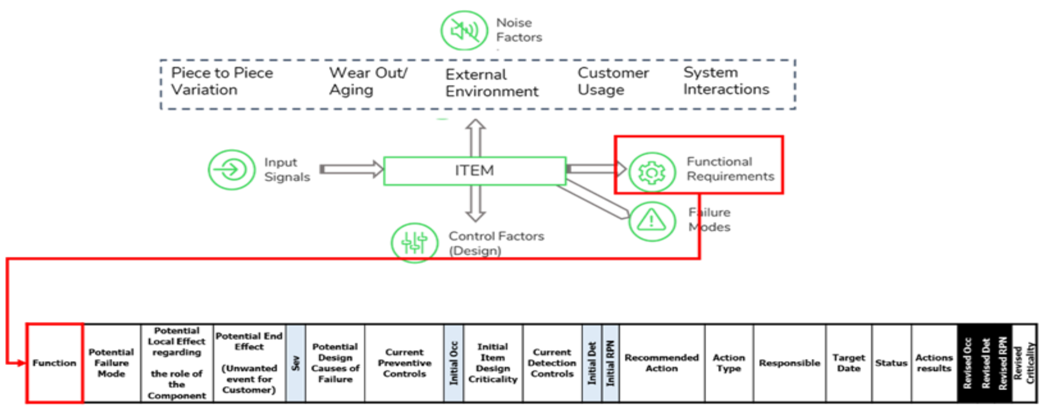


Figure 1 From P-Diagram to DFMEA - Function

2. POTENTIAL FAILURE MODE:

For each function entered, enter the potential failure modes associated with it from the P-Diagram. These failure modes could be in the form of:

- Total Loss of the Function
- Degradation of the performance of the Function
- Intermittent Function
- Unexpected behavior of the Function

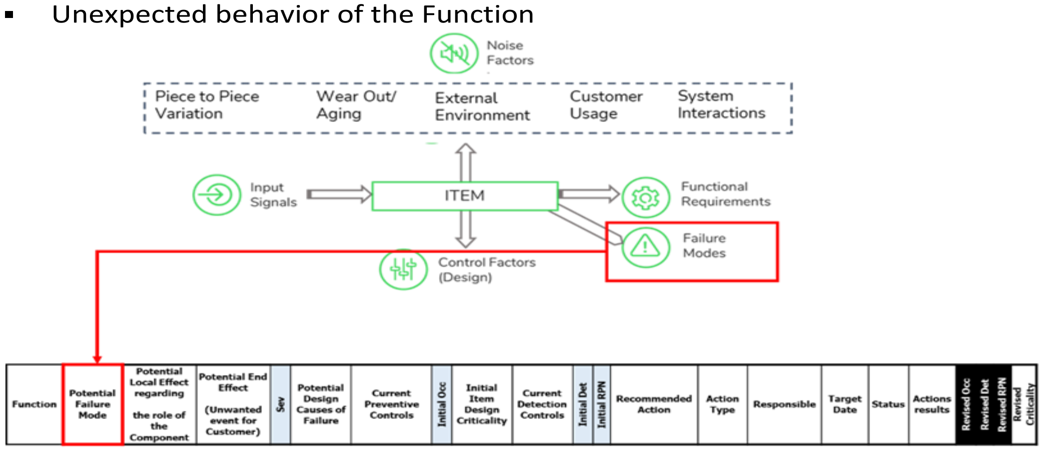


Figure 2 From P-Diagram to DFMEA - Failure Modes

A. Conducting the DFMEA (cont.)

3. POTENTIAL LOCAL & END EFFECT:

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 3 DFMEA - Potential Local Effects

Identify the local effects (how the failure mode impacts the component, subsystem or the system) AND the end effects (what the end customer will experience) as a result of each failure mode. End effects could be potential safety, compliance, property damage or customer dissatisfaction. In case of having more than one end effect the team should use the worst-case end effect. Severity **(SEV)** ranking is based on the end effect and NOT based on the local effect or based on the likelihood of occurrence. The DFMEA team shall utilize the latest controlled copy of End Effect tables from the Reliasoft XFMEA global instance, which is controlled by the LoB DfSR leader and Offer Safety Officer.

The list of standard End Effects provided by each LOB is also available in Appendixes A3 – A7 as an illustration only.

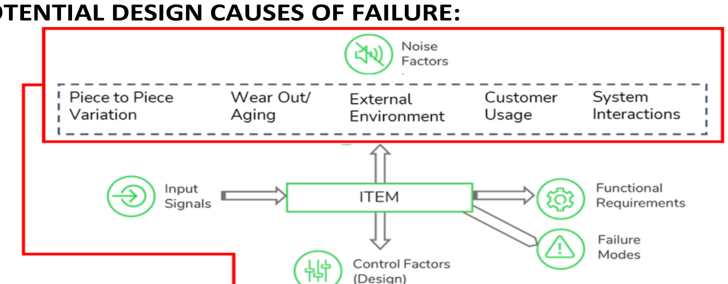
4. SEVERITY:

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 4 DFMEA – Severity

Enter the **Severity (Sev)** rating associated with the end effect from the standard Severity tables developed and approved for each LoB (**See Appendix A3, A4, A5, A6, and A7.**) These severity tables are already stored in Reliasoft for use. The generic Severity table is also provided in **Appendix A1.** Teams should always use the latest controlled copy of the Severity tables from Reliasoft.

5. POTENTIAL DESIGN CAUSES OF FAILURE:



Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 5 From P-Diagram to DFMEA - Noise Factors

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A. Conducting the DFMEA (cont.)

Enter the associated design causes for the failure mode from the P-Diagram. These causes originate from the five noise factors.

It is necessary to describe the noise factors / root cause as a physics-based design weakness. This will involve a minimum of two elements, and usually three as outlined below:

Element1 – Failure Mechanism: Describe the failure mechanism in terms of the physics, or the science of failure. **As an example**, Part broken or fractured is not deep enough. *High Cycle Fatigue*, or *Brittle Fracture* is the proper resolution.
Element2 - Stressor: Identify the stressor (e.g. vibration, humidity, etc.) that drives the failure. For example, *High Cycle Fatigue as a function of Pressure Cycling*.
Element 3 – Design Margin: Include the design feature that permits the failure mechanism to occur. For example, High Cycle fatigue as a function of pressure cycling AND *insufficient fatigue strength due to a sharp corner by design.*

This column should be used by design engineer to identify any design margin (dimensions, material properties, etc.) that could lead to potential safety issues, compliance risks, or property damage. and must be classified as **CTQ** and transferred to the drawings, **CTQ** list and control plans for that specific focus item.
Before identifying the "control factors / recommended actions" in the P-Diagram and DFMEA, we need to evaluate the RPN following the steps below.

6. CURRENT PREVENTIVE CONTROLS:

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 6 DFMEA - Current Preventive Controls

Enter the current preventive controls (planned and/or executed) for the associated design cause. They may include analytical activities that do not require any physical testing. Examples of preventive controls include:

- Design guides
- Analytical modeling, such as simulation and Finite Element Analysis (FEA), etc.

7. INITIAL OCCURRENCE:

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 7 DFMEA - Initial Occurrence

A. Conducting the DFMEA (cont.)

Rank the probability/likelihood of Occurrence (O) of the failure cause (**Appendix B1** and **Appendix B2**) based on the **current** preventive controls identified in the DFMEA, review the following questions from the Occurrence table to estimate the probability of occurrence of the failure cause:

- Does this cause/failure appear in any previous similar studies or product?
- What is the quality history on similar products?
- Is the selected solution new or old (well known)?
- The justifying elements let think that a potential risk remains.
- If the solution has been previously implemented; is its operational environment different, have new functions been added, have the constraints been changed from before or have the applicable offer requirements been modified since the design of the last product?

8. INITIAL ITEM DESIGN CRITICALITY:

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 8 DFMEA - Initial Item Design Criticality

Assess component criticality based on Severity (S) and Occurrence (O) Rating. Component criticality levels are shown in the Figure 9.

				SEVERITY		
		Imperceptible	Performance	Working	Damage	Safety
	Almost uncertain	Minor	Minor	Minor	Damage-Major	Safety-Major
OCCURRENCE	Low	Minor	Performance-Major	Working-Major	Damage-Critical	Safety-Critical
	Medium	Minor	Performance-Major	Working-Major	Damage-Critical	Safety-Critical
	Almost certain	Minor	Performance-Major	Working-Critical	Unacceptable	Unacceptable

Figure 9 Severity Table

9. CURRENT DETECTION CONTROLS:

List all current (executed and planned) verification and validation tests aimed at detecting the associated failure mode and design causes. These include all the physical tests (rig, bench, functional, environmental, reliability, etc.) carried out on future products (prototype or production unit.)

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 10 DFMEA - Current Detection Controls

10. INITIAL DETECTION

Detection Score (D) reflects the effectiveness of current design controls in detecting the *failure mode* caused by the associated *design causes*. Refer to the **Detection Rating Table in Appendix C1** and **Appendix C2** to assign the correct detection rating. The team must always use the **controlled copy of the Detection Ratings stored in ReliaSoft** to ensure consistency and compliance.

A. Conducting the DFMEA (cont.)

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 11 DFMEA - Initial Det Score

11. INITIAL RISK PRIORITY NUMBER (RPN)

Assess the potential design risk using Risk Priority Number (RPN) which is calculated as: RPN = SEV x OCC x DET

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 12 DFMEA - Initial RPN

The RPN is used to rank and prioritize the actions to mitigate the risk. The DFMEA team should develop an RPN matrix as shown in Figure 13 to manage the potential design risk.
RPN results can be visualized in a grid with defined rules on thresholds to launch actions.
The RPN helps in identifying and prioritizing the corrective actions to be put in place to reduce risks identified in the FMEA. In practice, and with scales of ranking for SEV, OCC and DET going up to 10:

- If RPN **green**: in general, no corrective action is necessary.
- If RPN **orange**: corrective actions should be considered.
- If RPN **red** or **dark**: one or several corrective actions must be taken in order to reduce the RPN to an acceptable level based on the criteria which is explained below.

		RPN = Severity * Occurrence * Detection			
Severity		RPN ≤ 64	70 ≤ RPN ≤ 112	144 ≤ RPN ≤ 343	360 ≤ RPN
Imperceptible	1				
Performance	4				
Working	7				
Compliance/ Property Damage	9				
Human Safety	10				

Figure 13 RPN Matrix

12. RECOMMENDED ACTION AND ACTION TYPE:

Team shall mitigate the potential design risk by identifying Recommended Actions:

- **Begin by looking at the highest Severities**. In particular, focus on severities of 9 and 10. if any severities of 9 or 10 exist (safety/regulation severities)

General

A. Conducting the DFMEA (cont.)

with a RPN of 144 or more it requires mitigation activities as the first priority. The team should start with the **dark black cell** (RPN>=360) first then move to the red cell. **Moreover, an FTA analysis is required for the failure modes with severity of 9 or 10** (Please refer to **GL-ProQ-018 Fault Tree Analysis (FTA) Procedure)**

- Next, investigate failures modes with **severity of 7** (potential Robustness and Reliability risk) starting with the Black Cell first and then moving to the Red cell.
- Next, the team continues to mitigation activities for the “Red Cells” for the Severity of 4.

The intent of the recommended actions is to reduce the potential risks with design mitigations only. For example, the test during production is not valid **Recommended Action** during DFMEA. Identifying and implementing these actions should consider reducing risk ratings **in the following order**:

- Design modifications to reduce occurrence (O) of the **failure cause**
- Improve (smaller the better) the detection (D) ability for the **failure cause or failure mode**

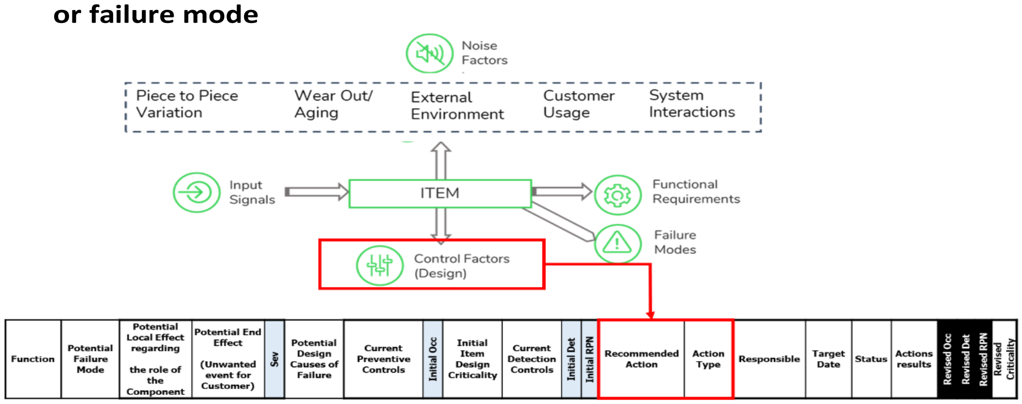

Figure 14 From P-Diagram to DFMEA - Control Factors

Select the **type of action by using**:

- **Design out:** Identified new design improvement actions to **eliminate or mitigate the failure end effect or the root causes of the failure.**
- **Guards:** Design-in controls or mechanisms that detect or limit the effect of a failure when it cannot be fully designed out. Examples include:
  - Enhancing detection of causes or mechanisms
  - Recognizing failure modes and initiating appropriate system response
  - For Sev = 9 or 10 (safety/regulatory): the system may be required to transition to a fail-safe state.
  - For Sev = 7 (robustness/reliability): a derated performance mode with user warning may be acceptable and sufficient.
- **Warn about:** recommendations to be integrated into the Product documentation or label to warn the product users.

The identified recommended actions can be populated from the control factors field in the P-Diagram.

A. Conducting the DFMEA (cont.)

13. RESPONSIBLE/TARGET DATE/STATUS:

Each action should have a responsible person and a target completion date associated with it as depicted in Figure 15.

Status of the action should be entered by the responsible person:

- Open
- Completed
- Not Implemented

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Sev	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 15 DFMEA - Recommended Actions

14. ACTION RESULT/RESULTING POTENTIAL RISK

The team should state the Action results and enter the expected Occurrence, Detection and RPN score based on the results of implemented actions. The team then should update the RPN scorecard.

Function	Potential Failure Mode	Potential Local Effect regarding the role of the Component	Potential End Effect (Unwanted event for Customer)	Severity	Potential Design Causes of Failure	Current Preventive Controls	Initial Occ	Initial Item Design Criticality	Current Detection Controls	Initial Det	Initial RPN	Recommended Action	Action Type	Responsible	Target Date	Status	Actions results	Revised Occ	Revised Det	Revised RPN	Revised Criticality

Figure 16 DFMEA - Action Result

B. Utilizing the Results of the DFMEA in driving other DfSR Tasks

When the DFMEA has been updated with the expected potential risk upon completion of all the approved actions, the DFMEA shall be used to support required DfSR activities in the following manner:

1. All analytical predictions, analyses and modeling as identified within the DFMEA should be identified in the “Verification” section of the item’s Design Verification Test (DVT) plan.
2. All functional/performance, regulatory, environmental and reliability testing as identified in the DFMEA should be identified in the “Validation” section of the item’s DVT Plan.
3. Relative to the above, an effective DFMEA should have a direct mapping backwards to the Interface Diagram/Matrix and P-Diagram, and a direct mapping forward to the DVT Plan.
4. In those key areas of safety, compliance and reliability risk (higher RPN), CTQ’s are to be identified on drawings. The design parameters (geometric, surface finish, material property, etc.) of which have influence on the item’s safety, compliance and reliability, and were identified in the DFMEA are strong candidates for CTQ’s to be controlled strictly.

C. Sustaining the DFMEA

The DFMEA is not to be considered a static document. It should be revisited in Reliasoft as follows:

1. As recommended actions are completed, update the DFMEA to reflect what was done, by whom, and when. Reassess occurrence and detection ratings accordingly. If residual risk remains high, additional mitigation actions may be necessary.
2. **Continuously update** the DFMEA with performance data from manufacturing and field operations. This includes updating failure modes, causes, and rating values to reflect actual product behavior.
3. When initiating a new DFMEA or P-Diagram, review existing entries in the ReliaSoft repository. Well-executed and relevant DFMEAs can serve as valuable starting points, saving time and enhancing quality.
4. The maintained and up-to-date DFMEA must serve as the **Single Source of Truth** for design risk management.

Process Outputs

The results of the DFMEA will be used as one of the criteria for making Go/No Go decisions for offer/solution (such as Launch Commitment).

- Initial and Final RPN Matrix
- Recurrent Action Tracker Meeting
- FTA for Severity 9s and 10s

DFMEA in OLM

The offer's /solution's system level DFMEA must be ready for review by Engage.
The fully developed DFMEA for System / Subsystem / Component levels must be ready for review by Launch Commitment.

Monitoring & Control

- Offer Life Cycle Management (OLM) The DFMEA development is integrated in the OLM process. System level DFMEA shall be ready for review by Engage; and Fully developed DFMEA shall be ready for review by Launch Commitment.
- Quality Management System (QMS) Audits: QMS will ensure compliance to DFMEA deliverables defined in OLM by performing regular audits at each of the LOBs.
- DfSR Maturity Assessment: Corporate DfSR function will perform annual DfSR Maturity assessment where maturity of DFMEA execution on new offers / solutions development will be assessed for each LOB and by each region.

Roles & Responsibilities

As DFMEA development is a cross-functional activity, a **RACI** matrix as described in table below should be followed. The team may need to consult additional members.

Function	RACI Roles
Design Engineer	R
Offer Technical Leader	A
Scrum Master	I
Industrialization Quality Leader (IQL)	C
DfSR Practitioner/Leader*	C
Test Engineer	C
Offer Quality Leader (OQL)	C
Offer Safety Officer (OSO)	C
Service Engineer	C

* DfSR Practitioner is an SME on DFMEA methodology and serves as a facilitator for DFMEA

General

Competency Requirements

Role	Knowledge	Competency Level	Intervention
DfSR Practitioner/Leader	Advanced P-Diagram Methodology & This procedure	Expert	Advanced DFMEA Practitioner Training
Design Engineer & Offer Technical Leader	This procedure	Advanced	Advanced DFMEA Practitioner Training
Industrialization Quality Leader (IQL)	This procedure	Competent	P-diagram Awareness Training
Offer Quality Leader (OQL)	This procedure	Competent	Advanced DFMEA Practitioner Training
Test Engineer	This procedure	Competent	Advanced DFMEA Practitioner Training
Offer Safety Officer	This procedure	Competent	Advanced DFMEA Practitioner Training
Scrum Master	This procedure	Basic	DFMEA Awareness Training
For Other Functions Not Listed	This procedure	Basic	DFMEA Awareness Training

Quality Records

Record	Record Location	Retention Period
Developed DFMEA	PCIM - Project Folders	20 years

Terms & Acronyms

The table below includes definitions for terms and acronyms used in this document. Reference **iSEE** for a full list.

Term	Definition
DFMEA	Design Failure Mode and Effects Analysis
DfSR	Design for Safety & Reliability
P-Diagram	Parameter Diagram.
DVT	Design Verification Test
FTA	Fault Tree Analysis
PPAP	Production Part Approval Process
RACI	Responsible, Accountable, Consulted, Informed
PRB	Problem Review Board.
SE	Schneider Electric
LOB	Line of Business
OLM	Offer Lifecycle Management
QMS	Quality Management System

Appendix A1: Severity Table for Hardware & Systems

S	Level	Failure Effect on (End) user / Customer
1	**Imperceptible**	**Undetectable by the Customer.** Potential failure effect: no perceptible effect on product functionality *Customer is unlikely aware of it.*
4	**Performance**	**Still working.** Potential failure effect: some product degradation at the subsystem level, but performance of main systems is not affected. *Customer is made uncomfortable or annoyed.*
7	**Working**	**On-site support needed.** Potential failure effect: inoperable subsystems or degradation of main systems, and main product functionality is lower than Customer expectations (for example: all CTQ coming from persona profile which severity level was primary ranked @ 1 or 4) *Customer dissatisfaction is experienced*
9	**Compliance/ Property damage**	**Major Customer business impact: destruction of part of installation or startup impossible.** Potential failure effect: non-conformance to regulation (risk of prosecution or huge cost) or inoperable product at the highest system levels, but is not safety related. *Customer is highly dissatisfied*
10	**Safety - Human injury**	**Potential safety issue only.** Potential failure effect: injury or harm to the health of human beings. This includes items of non-conformance to regulation representing direct potential safety problem. *Customer is endangered.*

Appendix A2: Severity Table for Software & Firmware

S	Level	Criteria for Firmware	Criteria for Software
1	**Imperceptible**	Minor cosmetic issues not affecting functionality or performance. No noticeable effect on device operation or user experience.	Insignificant visual glitches or minor bugs not affecting user operations. No impact on data accuracy or system performance.
4	**Performance**	Subsystem performance degradation not impacting main system operations. Minor firmware bugs causing slight delays or inefficiencies in non-critical functions.	Minor performance issues causing slight inconvenience to users. Software bugs leading to temporary slowdowns or minor functional restrictions.
7	**Working**	Firmware issues requiring technical support or manual intervention. Significant performance drops or intermittent failures in main system components.	Software malfunctions requiring user intervention or support. Significant disruptions in main functionalities requiring troubleshooting.
9	**Compliance/ Property damage**	Critical failures causing system downtime or damage to equipment. Non-compliance with regulatory standards leading to operational halts.	Major software errors causing data loss or severe operational disruptions. Non-compliance with legal or regulatory requirements leading to fines or operational delays.
10	**Safety - Human injury**	Firmware bugs posing risks of physical harm. Severe failures leading to hazardous conditions or regulatory breaches.	Software vulnerabilities causing safety hazards or data breaches. Critical issues leading to potential harm to users or severe legal consequences.

Appendix A3: Standardized Severity Tables for HS

Potential End Effect (Unwanted Events)- HS	Severity
Electrical Shock	10
Bodily Injury/Harm (other than Electrical Shock)	10
Property Damage	9
Environmental Harm	9
Compromised Regulatory Compliance	9
Compromised Sustainability	7
Branch Circuit Power Loss	7
Product Replacement	7
Delayed / Compromised Installation	7
Compromised / Degraded Energy Management or Control Capability	7
Compromised / Delayed Diagnostics and/or Maintenance	4
Cosmetic	1
No perceptible issue	1

Appendix A4: Standardized Severity Tables for PP

	Potential End Effect **Power Products** Devices	Severity	Category
[PP] (10)	Arc Flash Hazard	**10**	Safety – Human injury
[PP] (10)	Unintentional equipment operation (Closing)		
[PP] (10)	Release of Hazardous Materials of Substances (Toxins, Carcinogens, etc.)		
[PP] (10)	Non opening on protective device which can create Arc flash, electrical shock and/or fire hazard end effects		
[PP] (10)	Loss of phase/phase or phase/ground insultation		
[PP] (10)	Human factors envelop exceeded (short term / Long term bodily injury)		
[PP] (10)	Fire and burn hazard		
[PP] (10)	Explosion/ Expulsion		
[PP] (10)	Signalization defect : Gap between power status and auxiliary/signalization status (Open indicated but Close)		
[PP] (10)	Signalization defect : ERMS notified while not engaged		
[PP] (10)	Out of Process Safety Time (PST)		
[PP] (10)	Electric Shock Hazard		
[PP] (9)	Customer equipment damage (other than defective device)	**9**	Major Customer Property damage
[PP] (9)	Installation of a not certified Firmware		
[PP] (9)	Non-compliance to standards with Follow-up (UL, IEC or certification renewal) with property damage		
[PP] (7)	Visual aspect - perceived quality	**7**	Service continuity
[PP] (7)	Lack or Loss of traceability defined by the project		
[PP] (7)	Unintentional equipment operation (Opening or nuisance trip)		
[PP] (7)	Transportation product damage		
[PP] (7)	Signalization defect: Other cases		
[PP] (7)	Product early ageing		
[PP] (7)	Packaging damage with product damage		
[PP] (7)	Operating time longer than expected = Operation duration slower than expected		
[PP] (7)	Non opening on command/control device		
[PP] (7)	Non closing all devices		
[PP] (7)	Noise		
[PP] (7)	Improper fixation - Product fixation failure		
[PP] (7)	Green Premium compliance		
[PP] (7)	Equipment or product start/assembly impossible		
[PP] (7)	Loss of selectivity		
[PP] (7)	Additional Technical Expertise (primary function)		
[PP] (7)	Defined maintenance could not be operated		
[PP] (4)	Additional Technical Expertise (secondary function)	**4**	Performance
[PP] (4)	Loss of diagnostic		

Appendix A5: Standardized Severity Tables for PS

Potential End Effect (Unwanted Events)- PS	Severity
No tripping under fault (voltage or current)	10
CB stuck in intermediate position when opening	10
CB current path Over Heating	10
No tripping under fault (voltage or current) --> Melted Contacts	10
CB current path Over Heating	10
No tripping under fault (voltage or current)	10
Jeopardize at least one main function	10
No tripping under fault (voltage or current)	10
Arms misalignment	10
Opening speed over the upper speed limit	10
loss of insulation	10
Fire propagation in the cubicle	10
Wrong status: Displayed Opened when actually closed	10
CB stuck in racked in position: Sectioning not possible	10
Access to live parts	10
CB Toppling during handling	10
Loss of electrical continuity in the CB	7
Impossible to close CB	7
CB stuck in racked out position	7
Impossible to open CB	7
Tripping time longer than selectivity setting --> Loss of service continuity	7
Spurious / Nuisance trip (tripping when no fault)	7
Opening speed below the lower speed limit	4

Appendix A6: Standardized Severity Tables for IA

Potential End Effect (Unwanted Events)- IA	Severity	Comments
[IA] (10) Unintended Equipment Behaviour Causing Bodily Damage	10	Could be caused by software bugs leading to unexpected movements, activations, or deactivations causing human injury or death. Example: A robotic arm moving unexpectedly due to a firmware error.
[IA] (10) Arc Flash	10	Software controlling high-voltage switching could malfunction, leading to an arc flash. Firmware controlling protective relays could fail to trip in time.
[IA] (10) Electrical Shock	10	Software controlling insulation monitoring or safety interlocks could fail, creating a shock hazard. Firmware controlling power supplies could malfunction.
[IA] (10) Fire Hazard	10	Overheating due to software mismanaging thermal control systems. Firmware errors in power regulation could cause overheating or short circuits.
[IA] (10) Customer Bodily Damage or disease (Short & Long Term)	10	Injury like cutting, crushing, burn. Exposure to RF, Electromagnetic waves, exposure to Radioactive, Carcinogens or Toxic substances in low or high amounts. Attack by fauna, vermin, microbes hosted on or in the product (due to lack of sealing of entry points). Software controlling safety systems (e.g., radiation shielding, chemical containment) could fail. Firmware could provide incorrect data related to hazardous conditions.
[IA] (9) Unintended Equipment Behaviour Causing Property Damage	9	Like unexpected start of movement causing shock with customer machine. Software bugs causing unexpected movements or activations leading to collisions or damage. Firmware related to motor control or actuator positioning could be a factor.
[IA] (9) Customer Equipment Damage	9	Degradation of customer equipment by heat, water Software managing resource allocation (e.g., power, cooling) could malfunction, leading to overheating or other damage. Firmware could send incorrect signals.
[IA] (9) Customer risk of prosecution, financial loss due to incompliance to regulations	9	Software or firmware not meeting regulatory requirements (e.g., safety standards, data privacy). Example: Data logging software not compliant with GDPR.
[IA] (9) Customer image compromised, loss of business	9	Non-Compliance with Environmental Policy, ethical, repairability, cyber security. Software vulnerabilities can lead to data breaches or cyberattacks.
[IA] (7) Unexpected Interruption of Customer's Operations	7	Product stops working. Software crashes, bugs, or performance issues. Example: A manufacturing line stopping due to a PLC firmware error.
[IA] (7) Financial Loss (Loss of Productivity)	7	Product works at lower performance, Unable to meet production rates, unable to deliver. Software inefficiencies or bugs reducing performance.

[IA] (7) Customer Process Availability Degradation (Service Life Reduced)	7	Customer process not running at designed levels due to sub-standard product performance. Software managing wear levelling or maintenance functions could be faulty.
[IA] (4) Bad Perceived Quality, Loss of Confidence in Product	4	Product works fine but aesthetics is poor, or noise or other subjective factors (i.e.: material, surface finish, etc.) Software controlling user interfaces could be poorly designed or buggy.
[IA] (4) Process interruption caused by False Alarm	4	Product triggers an alarm, causing customer to maintenance when it is not needed. Software algorithms for alarm detection could be poorly calibrated.
[IA] (4) Increased Total Cost of Ownership	4	Higher cost of maintenance, early replacement cycle, Premature aging, Reduced Useful Life. Software updates requiring significant downtime or complex procedures.
[IA] (4) Customer Annoyance	4	Nuisance Tripping, quick workaround available, almost no loss of productivity, Not easy to install, not intuitive installation or prompts. Poorly designed software user interfaces or confusing error messages.
[IA] (1) No perceptible Effect	1	Minor defects. Minor software bugs with no noticeable impact on functionality or cosmetic glitches in firmware-driven displays.

Appendix A7: Standardized Severity Tables for SP

Potential End Effect (Unwanted Events)- SP	Severity	Comments
[SP] (10) Arc Flash Hazard	10	Arc Flash in the frame not stop fast enough by protecting devices Arc Flash hazard in E-box or electrical wiring that is exposed to field team
[SP] (10) Release of Hazardous Materials/ Chemicals	10	Battery acid leak with risk of skin burn Droplet of electrolyte on Li ion battery Hot oil or refrigerant burn to skin
[SP] (10) Fire/Flame & burn hazard	10	Fire going outside of the enclosure Hot surface with potential for injury with no warning. Battery acid leak with short to chassis and risk of fire
[SP] (10) Explosion/ Expulsion hazard	10	Projection from pressurized gas Door / panel being projected.
[SP] (10) Physical Injuries	10	Cuts from sharp edges Any issue leading to a medically confirm bodily injury
[SP] (10) Electric Shock Hazard	10	IP2X is breached and harmful voltage can be touched by user. IPX1+ (project with some level of water resistance) is breached with water leading to harmful voltage Signalization defect leading to inaccurate identification of L, N and GND Signalization defect leading to unexpected harmful voltage (product off but harmful voltage available on the output) Grounding continuity issue (rack with grounding discontinuity)
[SP] (9) Customer equipment damage (other than SE defective device)	9	Other equipment damaged in the rack due to SE product
[SP] (9) System destruction (non repairable)	9	A fault in a subsystem propagates and lead to destruction of adjacent subsystems. Repair costs significantly increased. Customer is highly dissatisfied. Risk of law suit.
[SP] (9) Non compliance to external standards (UL, IEC,...)	9	External standard not met (And not related to Safety)
[SP] (9) Unintended Load drop on Redundant system	9	UPS control system issue leading to unexpected output off and load drop on a parallel/redundant system.
[SP] (7) Onsite support needed	7	A fault that require Schneider FSR to go to site (taken from FTD00888)
[SP] (7) Loss of main functionality	7	IPX1 breached (product doesn't meet its IPX1 or more specification without risk of water intrusion in section but no harmful voltage is available) Display issue leading to full product replacement (small systems)

[SP] (7) Product doesn't meet specification	7	Noise above specification Product does not meet design intention. Example - twin cool unit does not have capability to automatically switch from DX to Water mode Green Premium compliance
[SP] (7) Unintented load drop non-redundant system	7	UPS brain issue leading to unexpected output off and load drop Output interruption / short power losses (due to IGBT failure, Capacitor failure, fuse open...)
[SP] (7) Unintented reduction of output power / load protection non-redundant system	7	Power Module failure and system is operating with reduced output UPS goes to bypass due to a UPS issue (IGBT failure, Capacitor failure, fuse open...)
[SP] (7) Unintented reduction of output power / load protection redundant system	7	Loss on single cabinet in parallel system Forced bypass in parallel/redundant system
[SP] (7) Reduced UPS battery backup time	7	Back-up time doesn't meet the communicated information with new battery Back-up time doesn't meet the communicated information (e.g. remaining capacity does not meet 95% of inital capacity after warranty time) Real back-up time lower than UPS prediction Incorrect battery status/health indication
[SP] (7) Damaged product	7	Damage product / package rejected by customer (1Ph)
[SP] (7) Product can't be operated	7	Dead on arrival - product can't be commissioned or is rejected for quality reason by customer Product doesn't starts anymore Battery is swollen bulge and can't be operated anymore (without Safety risk) Lead acid Battery failure leading to release of gases (Rotten smell) & customer is not willing to operate with existing battery Critical alarm generated to display
[SP] (4) Warning alarm (without loss or reduction of output)	4	Fan alarm but product is still able to operate fully Display issue but output is still on (and display as spare part - large systems) Warning alarm generated to display
[SP] (4) Minor Non-conformity like visual defects	4	Packaging minor damage Damage product / package rejected by customer (3Ph) Minor scratches
[SP] (4) Offer doesn't meet a secondary functionality (performance) specification/parameters	4	Issue with humidity/temperature sensor for ambient environment info UPS detects end of life of the lead acid battery slightly before expected battery end life

[SP] (4) Loss of redundant power/inverter module in a modular/redundant system	4	A power module in a modular/redundant system shuts down due to a fault. The fault is kept isolated to the power module. Remaining modules in the system take over the load. Customer/end user can balance system capacity and load, so that the loss of the power module can be tolerated without overloading full system. With Live Swap, a QUALIFIED PERSON may replace the module "on the fly".
[SP] (4) Loss of air fan (or equivalent subsystem) with risk mitigation	4	An air fan (or equivalent subsystem) is lost and system responds to compensate for it (i.e. by spinning up remaining fans and/or shift load away). A QUALIFIED PERSON may replace the fan (or equivalent subsystem) "on the fly".
[SP] (4) Loss of a redundant system PSU in system with redundant PSU's	4	Redundant PSU outlets to takeover all internal/external supply needs. With Live Swap, a QUALIFIED PERSON may replace "on the fly".
[SP] (4) Loss of redundant UPS brain in system with redundant controller	4	UPS brain issue leading to takeover by backup redundant controller. With Live Swap, a QUALIFIED PERSON may replace "on the fly".
[SP] (1) No impact on customer application, not visible	1	

Schneider Electric

Appendix B1: Occurrence Table for Hardware and Systems

Occ	Level	Design Ranking Criteria	ELM Design Real Examples	ELN Design Examples
1	**Almost uncertain**	• New Design Demonstration with design fundamentals including simulation has been performed to withstand **the characteristics** **and** Peer review organized with no weak point identified, **and** Demonstration done within reliability demonstration, **and** Demonstration follow internal design rules. • Re-use of brick or design in same conditions already demonstrate reliability with a clear risk assessment, or Return rate is compliant to Offer Quality Performance objectives	Design (**mechanism)** validated with relevant numerical simulation on model already validated on existing brick	Apply Schneider Electric internal design rules (component sizing, peer review, worst case analysis, parameter degradation analysis) Multiple experiences of satisfying usage of this type of component in same conditions, proven at similar reliability level as targeted, by trustable field experience or qualification in application.
4	**Low**	• Demonstration based on legacy design or field experience • Re-use of brick or design in same conditions already demonstrate reliability without clear risk assessment, or Low Customer returns • Based on know-how without demonstration	change of raw material and re-use another one: "re-use of brick... w/o clear risk assessment"	similar design but change of component or reuse of brick w/o clear risk assessment The component, except inappropriate supplier / series, may generally fit with the targeted mission profile and reliability level, as known from previous trustable field experience or the state-of-the art.
7	**Medium**	• Calculation demonstrates weak characteristics achievement (design margin missing) • Re-use of technology/brick with partial control of compliance to mission profile, or with known weaknesses in same usage conditions • Design activity verification done only with product test, without unitary verification	- Lack of integration: sub-assemblies not well specified and **characterized** - For mechanisms: No dimension chains calculation	No design margin The component, depending on the supplier or series, may be inadequate for the targeted mission profile and reliability level, as known from previous failure experiences or the state-of-the art
10	**Almost certain**	• One product failure without root cause identified and mitigated, • No design margin identified • Application of technology, new in our operating experience nor No product verification and/or validation experience • Calculation demonstrate characteristics is not achieved • Re-use of technology/brick without controlling compliance to mission profile (potential performance weaknesses or Customer returns) • No demonstration with design fundamentals including simulation • Lack of technical need definition	- test failure isn't taken into account in design activities; - No identification of critical **characteristics**, not guaranteed by the supplier - For complex mechanisms (lot of Functional Conditions): No dimension chains calculation	New use of a component without any experience Component characteristics is defined @25°C but not @70°C, which is the technical need Component selected does not fulfill the requested Critical characteristics. Evidence of mismatch between this type of component and the targeted mission profile and reliability level, as known from previous failure experiences or the state-of-the art.

Appendix B2: Occurrence Table for Firmware & Software

OCC	Level	Short Description	Criteria for Firmware	Criteria for Software
1	**Almost uncertain**	Highly unlikely due to robust design processes and rigorous testing.	- Comprehensive simulation and modeling. - Thorough peer reviews of code. - Rigorous validation and testing processes.	- Robust design with peer-reviewed algorithms. - Extensive unit, integration, and regression testing. - Strict code review processes.
4	**Low**	Possible but not frequent, relying on past experiences and legacy designs without rigorous current validation.	- Maintenance of legacy protocols with infrequent updates. - Standard reliability checks. - Basic validation efforts.	- Reliance on legacy systems with periodic updates. - Basic validation and testing practices. - Adequate but not extensive memory management.
7	**Medium**	Notable weaknesses in design, with moderate likelihood of failure due to insufficient testing and validation controlS.	- Weak handling of edge cases. - Moderate testing procedures. - Limited error recovery and robustness.	- Design weaknesses with insufficient data validity checks. - Moderate testing and validation coverage. - Limited error handling and recovery mechanisms.
10	**Almost certain**	High likelihood of failure due to significant gaps in design, testing, and validation processes.	- Major design gaps with inadequate testing. - Insufficient validation, especially under high-load conditions. - Frequent issues and failures.	’ - Significant design flaws with minimal testing. - Poor validation processes. - High probability of failure under stress conditions.

Appendix C1: Detection Table for Hardware and Systems

DET	Level	Design Ranking Criteria	ELM Design Real Examples	ELN Design examples
1	Almost certain	**Detection almost certain.** - Proven comprehensive test method for verification of functionality and performance, quality, reliability and durability - Prior testing confirmed that failure mode or cause cannot occur, or detection methods proven to always detect the failure mode or failure cause. - Planned timing is sufficient to modify production tools before release for production. *Customer will unlikely face it.*	- A comprehensive model **exists**, with calibration, hypothesis under control by identification/specification of all major related characteristics.	functional verification by simulation (ex: PSPICE) AND by physical test at component AND sub assembly levels covering all usage conditions (temperature, variability,...) Full component source qualification done in application context as defined by CSRM process
4	Medium	**Medium chance of detection.** - Proven partial test method for verification of functionality and performance, quality, reliability and durability; - Planned timing is later in the product development cycle such that test failures may result in production delays for re-design and/or re-tooling. *Customer will sometimes face it*	- A model **exists**, but perhaps with some gap with reality, with **hypothesis** unclear or not under control. - Tests are conducted "at the nominal", without Design of Experiment approach or other technique. - A model doesn't **exist** but we have design/production history which demonstrate a quite good level of detection	component ageing test: based on ageing model from state of the art, and with standard sampling. - functional verification by simulation (ex: PSPICE) AND by physical test at sub assembly level covering all usage conditions (temperature, variability,...) - no physical test at component level covering all usage conditions (temperature, variability,...) Component source object of deep datasheet analysis and functional test in product verification context (white box tests)
7	Low	**Low chance of detection.** - Test method not designed specifically to detect failure mode or cause - New test method; not proven. *Customer will often face it.*	If exist, only partial simulation. Tests are **conducted** "at the nominal", without DOE or other technique. A model doesn't **exist** but we have design/production history which demonstrates not rare lacks in design issue detection.	- component ageing test not performed and compliance to mission profile deducted from similar tests performed on similar components for similar conditions - functional verification by simulation (ex: PSPICE) OR by physical test at sub assembly level covering nominal case Component source only tested through the product validation (black box tests)
10	Almost uncertain	**Detection almost uncertain.** Test procedure yet to be developed. Functionality and/or Performance is NOT tested or cannot be tested *Customer will certainly face it.*	- no test defined in validation plan - re-use of existing test results w/o new mission profile consideration	- No component ageing test performed, and no commitment from the supplier on compliance of the component to mission profile. - no functional verification by simulation (ex: PSPICE) OR by physical test at component OR sub assembly levels covering all usage conditions (temperature, variability,...) Component source only selected in CSM without datasheet analysis, and without test in product prototypes

Appendix C2: Detection Table for Firmware & Software

DET	Level	Short Description	Criteria for Firmware	Criteria for Software
1	**Almost certain**	**Detection almost certain.** *Comprehensive tests verify functionality, performance, quality, reliability, and durability Customer will unlikely face it.*	Incorporation of watchdog timers, mechanisms for graceful degradation, and robust fallback procedures to maintain operational stability under fault conditions.	Use of advanced error detection algorithms, automated self-tests, and real-time fault tolerance mechanisms to ensure early detection and mitigation of issues.
4	**Medium**	**Medium chance of detection.** *Partial test methods verify functionality and performance, though delays may occur. Customer will sometimes face it*	Regular automated testing of firmware, performance validation under various conditions, and implementation of comprehensive safety checks to ensure reliable operation.	Implementation of extensive validation tests, periodic static code analysis, and routine automated testing to catch potential issues during the development phase
7	**Low**	**Low chance of detection.** *Test methods are not specifically designed for failure modes and are unproven. Customer will often face it.*	Maintenance of error logs, execution of diagnostic checks, and scheduling of firmware updates to identify and rectify issues that may arise during operation.	Basic error indicators within the GUI, continuous monitoring features, and occasional validation tests to identify and address errors that were not caught in earlier stages.
10	**Almost uncertain**	**Detection almost uncertain.** Test procedures undeveloped, functionality, and performance untested. Customer will certainly face it.	Extensive prototype testing and limited scope trials to gather performance data and identify potential failure points, with minimal initial error detection measures.	Reliance on manual testing protocols and feedback from prototype trials, with minimal automated error detection capabilities, leading to a higher likelihood of undetected issues.

REVISION HISTORY SUMMARY

Approvers	Requested By	Change Description	Reason for Change	Revision	Revision Date	Contributors
ERIC JACQUOT; OMAR GHANI	YUNWEI HU	• Specified that Control Factors in P-Diagram should be populated only after the RPN ranking is completed. • Added a statement emphasizing that the maintained, up-to-date DFMEA must serve as the single source of truth for design risk management throughout the product development lifecycle. • Made additional wording improvements for clarity and consistency.	• Improve process clarity and sequencing. • Strengthen risk management governance: Designates DFMEA as the single, authoritative source for design risk information, preventing inconsistencies across documents and tools	2.0	02/02/2026	
ERIC JACQUOT; YAVUZ GOKTAS	YUNWEI HU	• Enhanced Root Cause Analysis Guidance • Clarified Action Descriptions • Updated Appendix A3	To Improve the clarity of the P-Diagram and DFMEA development procedure.	1.3	07/02/2025	
YAVUZ GOKTAS; ERIC JACQUOT	YUNWEI HU	* This document has been migrated from the Global QMS Documents Platform (SharePoint) without revision changes. Process owner/Document author and revision details are specified in the document itself. * A summary of previous revisions is available only in the migrated version.	Document migration	1.2	03/03/2025	